

Beyond Antithetic Variates: Benchmarking Variance Reduction for Monte Carlo Pricing of Barrier and Asian Options

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Abstract

The base paper prices an up-and-in barrier call and a geometric-average Asian option by Monte Carlo under geometric Brownian motion with antithetic variates, and names advanced variance reduction as future work. We rebuild its experiment from scratch, validate every estimator against closed-form prices, and reproduce its two central findings. Its antithetic speedups are confirmed as variance ratios (1.47× barrier, 1.35× Asian call) and shown to follow from the pair correlation ρ through $1/(1 + \rho)$. Its finding that the discretization scheme barely matters is confirmed and explained: Milstein’s pathwise gain (strong order 0.97 vs. 0.49) does not reach the price, where both schemes are first order. We then add three advanced techniques — control variates, randomized quasi-Monte Carlo (RQMC) with a Brownian bridge, and importance sampling — and price the arithmetic Asian option, which has no closed form. All five methods are benchmarked in one interleaved run on efficiency, $1/(\text{variance} \times \text{time})$, with bootstrap confidence intervals. At $N = 65,536$, control variates and RQMC are 10^2 – 10^3 times more efficient than plain Monte Carlo on the near-the-money barrier and both Asians; importance sampling specializes in rare knock-ins ($\approx 49\times$ at $B = 160$); antithetic variates give a modest 1.1–1.5×. Variance reduction also exposes a -0.011 discrete-monitoring bias that plain Monte Carlo cannot see. We recommend RQMC with a Brownian bridge for barrier options and the geometric-Asian control variate for the arithmetic Asian.

Keywords

Monte Carlo simulation, variance reduction, option pricing, barrier options, Asian options, control variates, quasi-Monte Carlo, Brownian bridge, importance sampling

Group and Section

Group 07 Section A

Code: <https://github.com/sleepytmzd/Numerical-project-G07>

1 Introduction and Base Paper Review

1.1 Motivation

Monte Carlo (MC) simulation is the dominant computational tool for pricing path-dependent exotic options, because it needs no assumption beyond the ability to simulate the underlying stochastic process. Unlike closed-form solutions, which exist only for special payoff structures, MC can price any claim whose payoff is a function of the simulated path. The cost is statistical: a crude MC estimate converges at the slow rate $O(N^{-1/2})$, so reducing an estimator’s variance without introducing bias is one of the most practically important problems in computational finance.

1.2 Background

A European option gives its holder the right, not the obligation, to buy (call) or sell (put) an asset at a fixed strike price K at maturity T . The Black–Scholes model [1] assumes the asset follows geometric Brownian motion (GBM) under the risk-neutral measure,

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad (1)$$

where S_t is the asset price, r the risk-free rate, σ the volatility, and W_t a standard Brownian motion. *Exotic options* are path-dependent: their payoff depends on the whole trajectory of S_t , not just its terminal value.

- A **barrier option** activates (“knock-in”) or extinguishes (“knock-out”) if the path touches a level B during the option’s life. An up-and-in call pays $(S_T - K)^+ \cdot 1_{\{\max_{0 \leq t \leq T} S_t \geq B\}}$. Reiner and Rubinstein [14] give a closed form under *continuous* monitoring; discrete monitoring introduces a downward bias that Broadie, Glasserman and Kou [3] quantify with a continuity correction (“BGK”).
- An **Asian option** pays on the average price. The geometric-average Asian has a closed form [9], because the geometric mean of lognormal variables is itself lognormal. The arithmetic-average Asian has no closed form, which is why it is the standard example for demonstrating that Monte Carlo is indispensable.

Exact GBM sampling at a step Δt uses

$$S_{t+\Delta t} = S_t \exp\left((r - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}Z\right), \quad Z \sim N(0, 1), \quad (2)$$

which has zero discretization error at the grid points for any Δt . Two alternative SDE discretizations are commonly used when no

exact scheme is available: Euler–Maruyama (strong order 0.5) and Milstein [11] (strong order 1.0, adding the Itô correction $\frac{1}{2}\sigma^2 S_t(Z^2 - 1)\Delta t$). Both have *weak* order 1, so for a price — which is an expectation — the scheme choice changes the answer only at $O(\Delta t)$, far below Monte Carlo sampling error at any practical N . Standard references for the following material are Glasserman [6] and Boyle, Broadie and Glasserman [2]. Four variance-reduction techniques exploit different structure to beat the $O(N^{-1/2})$ wall:

- (1) **Antithetic variates** pair each path driven by Z with a mirror path driven by $-Z$. The pair average cancels the part of the payoff that is linear (odd) in Z and does nothing for the even part, so the gain depends on how close to linear the payoff is.
- (2) **Control variates** subtract a regression-adjusted residual of a correlated quantity with a known expectation. The optimal variance-reduction factor is $1/(1 - \rho^2)$, where ρ is the correlation between the target and the control.
- (3) **Quasi-Monte Carlo (QMC)** replaces pseudo-random points with a low-discrepancy sequence (Sobol [8, 15]) that fills the sampling space more evenly, at rates up to $O(N^{-1})$ for smooth integrands of low effective dimension [2, 12].
- (4) **Importance sampling (IS)** simulates under a shifted measure so that rare but important events occur more often, then corrects with a likelihood ratio.

1.3 Base Paper Review

The base paper [7] prices an up-and-in barrier call ($S_0 = 100$, $K = 105$, $B = 110.6772$, $\sigma = 0.2$, $r = 0.03$, $T = 1$ year) and a geometric-average Asian call and put, including a variant averaged over the last 30 days, under three path-generation schemes (“Euler”, “Euler–Maruyama”, “Milstein”), with antithetic variates as its variance-reduction technique, and presents its results as confidence-interval plots against the number of paths. Where the paper leaves implementation details open (the number of time steps, the random seed, and the Asian strike and averaging grid), we fix them at the most plausible values: 252 daily steps and $K = 105$, the strike consistent with its figures. Its two central findings are:

- (1) antithetic variates accelerate convergence “by 1.5 times for Barrier options and 1.3 times for Asian options”;
- (2) the choice of discretization scheme has a negligible effect on accuracy.

It recommends that future work prioritize “advanced variance reduction techniques” over new path-generation schemes. Section 4.1 reproduces both findings and explains where each comes from. The paper’s scope is options with a closed form and a single variance-reduction technique; pricing an option without a closed form, and comparing techniques on an efficiency metric that accounts for the extra cost each one introduces, are natural next steps that we take here.

1.4 Our Extension

We take up the paper’s own future-work direction. Table 1 summarizes what we add on top of the base paper.

Table 1: What the base paper did versus what we add.

Aspect	Base paper	Our extension
Options	Barrier call; geometric Asian call/put, 30-day call	+ Arithmetic Asian (no closed form); deep ($B=140$) and rare ($B=160$) barriers
Variance reduction	Antithetic only	+ Control variates, QMC with a Brownian bridge, importance sampling
Antithetic gain	$\approx 1.5\times$ barrier, $\approx 1.3\times$ Asian	Reproduced ($1.47\times$, $1.35\times$); predicted by $1/(1 + \rho)$
Efficiency metric	Convergence speedup	$1/(\text{var} \times \text{time})$ with bootstrap CI; time-to-accuracy
Error metric	CI plots	RMSE against a 252-date reference price
Scheme comparison	Three schemes; effect negligible	Reproduced; explained by measured strong order ($0.49 / 0.97$) and weak order (≈ 1 for both)
Convergence rate	CI plots vs. N	Plain MC $\approx N^{-1/2}$; RQMC slopes -0.60 to -0.80
Averaging window	30-day geometric Asian	Window sensitivity quantified: 6.708 (30 dates) vs. 3.000 (full year)
Barrier bias	Mentioned, not applied	Quantified (-0.011 paper, -0.160 deep, at 252 dates); BGK correction
Rare events	$B = 110.68$ only	$B = 140$: IS $8.6\times$; $B = 160$: IS $\approx 49\times$

2 Methodology

2.1 Simulation Engine

A single path-generation routine, `generate_paths`, implements the exact scheme (Eq. 2) and the Euler–Maruyama and Milstein schemes on a uniform grid of `n_steps` steps. It never draws its own randomness: it takes a pre-built array of standard normals of shape $(n_{\text{paths}}, n_{\text{steps}})$ and an optional drift override. This single frozen interface is what lets every later technique reuse the same engine:

- **Antithetic variates** build the normals array as $[Z, -Z]$.
- **QMC** builds the normals from a scrambled Sobol sequence via the inverse normal CDF, then applies the Brownian bridge as an orthogonal linear map before the array reaches the engine (Section 2.3).
- **Importance sampling** passes a shifted drift $\mu = r + \sigma\theta$ through the drift-override parameter, simulating under the tilted measure while using the *same* normals and the same

discounting; the likelihood-ratio correction is applied afterward by the estimator, not by the engine.

All three schemes are fully vectorized. Every estimator returns a common record: price, standard error, 95% confidence interval, runtime, and an extra dictionary for technique-specific diagnostics (e.g. ρ and $\hat{b}$ for control variates, effective sample size for importance sampling). A generic sweep routine loops over (scenario $\times$ scheme $\times$ method $\times$ $N \times$ replicate), logs one row per replicate, and a shared benchmarking module aggregates rows into per-cell statistics: mean price, variance *across replicates*, bias against a reference price, $\text{RMSE} = \sqrt{\text{bias}^2 + \text{variance}}$, mean runtime, and efficiency. All randomness is seeded deterministically from one base seed, so every run is exactly reproducible.

2.2 Control Variates

For a discounted target payoff Y , a discounted control payoff X with known expectation μ_X , the control-variate estimator is

$$\hat{V}_{\text{CV}} = \frac{1}{N} \sum_{i=1}^N \left[Y_i - \hat{b}(X_i - \mu_X) \right], \quad \hat{b} = \frac{\widehat{\text{Cov}}(Y, X)}{\widehat{\text{Var}}(X)}. \quad (3)$$

At the optimal coefficient $b^* = \text{Cov}(Y, X)/\text{Var}(X)$, the theoretical variance-reduction factor (VRF) is

$$\text{VRF} = \frac{\text{Var}(Y)}{\text{Var}(Y - b^*(X - \mu_X))} = \frac{1}{1 - \rho_{YX}^2}. \quad (4)$$

We fit $\hat{b}$ on an *independent pilot sample* and apply it to a separate production sample, keeping the estimator exactly unbiased. We use the discrete geometric-average Asian as the control for the arithmetic Asian, and the terminal European call (Black–Scholes price) as the control for the up-and-in barrier.

2.3 Quasi-Monte Carlo with a Brownian Bridge

The Koksma–Hlawka inequality bounds the QMC integration error by $V(f) D^*(P_N)$, the variation of the integrand times the star discrepancy of the point set, and $D^*(P_N) = O((\log N)^d/N)$ for a Sobol net. In $d = 252$ dimensions the bound itself is useless; fast rates appear only when the integrand has low *effective* dimension and is smooth. A kink ($\max(\cdot, 0)$) slows convergence, and a jump whose boundary is not aligned with the coordinate axes, such as the knock-in indicator $1\{\max_j S_{t_j} \geq B\}$, slows it further.

All Sobol sequences are generated with SciPy’s quasi-Monte Carlo module [16]. A scrambled Sobol sequence [13] is balanced only at sample sizes that are powers of two; drawing any other N silently degrades it toward the Monte Carlo rate, so our sample grid is restricted to powers of two and the code rejects any other N . Uniform points are mapped to normals only through the inverse normal CDF, never Box–Muller, which would mix coordinates and destroy the low-discrepancy structure. SciPy’s points are multiples of 2^{-30} , so an exact 0 (with $\Phi^{-1}(0) = -\infty$) has probability 2^{-30} per coordinate; each point is moved to the centre of its dyadic cell first, which keeps every normal finite across the $\sim 4 \times 10^8$ coordinates of a sweep.

A Sobol sequence is only genuinely low-discrepancy in its first few dimensions. Our path has 252 time steps, i.e. 252 dimensions, so assigning dimension j to time step j (the “incremental” construction) wastes most of the sequence’s advantage on high-index

dimensions where it is no better than random. The **Brownian bridge** [4] instead builds the path outside-in: dimension 0 sets the terminal value W_T , dimension 1 sets the midpoint $W_{T/2}$ conditional on W_T , and later dimensions fill in successive midpoints by recursive bisection,

$$W_t \mid W_{t_l}, W_{t_r} \sim \mathcal{N}\left(\frac{(t_r - t)W_{t_l} + (t - t_l)W_{t_r}}{t_r - t_l}, \frac{(t - t_l)(t_r - t)}{t_r - t_l}\right), \quad t_l < t < t_r. \quad (5)$$

This is an orthogonal linear map of the input normals, so the law of the simulated path is unchanged; only which input dimension drives which feature of the path changes. Randomized QMC (RQMC) draws R independent scrambles of the same Sobol net; this is the *only* valid source of error bars for QMC, since the points within one scramble are not independent. Each scramble is unbiased, so our RQMC estimator returns one price per scramble and logs its within-run standard error and confidence interval as NaN by design; every RQMC error bar in this paper comes from the spread across the R scrambles.

2.4 Discretization Order

For $dS_t = rS_t dt + \sigma S_t dW_t$ with step Δt and $Z_j \sim N(0, 1)$, the two discretizations are

$$S_{j+1}^{\text{EM}} = S_j(1 + r\Delta t + \sigma\sqrt{\Delta t} Z_j), \quad (6)$$

$$S_{j+1}^{\text{Mil}} = S_j(1 + r\Delta t + \sigma\sqrt{\Delta t} Z_j + \frac{1}{2}\sigma^2\Delta t(Z_j^2 - 1)), \quad (7)$$

against the exact step of Eq. 2, which uses the same Z_j . We measure the *strong* (pathwise) error $E|S_T^{\Delta t} - S_T|$ and the *weak* (price) error $E[f(S_T^{\Delta t})] - E[f(S)]$ for a payoff f , and fit their orders as log–log slopes against Δt . Theory gives strong order 0.5 (Euler–Maruyama) and 1.0 (Milstein), and weak order 1 for both. A price is an expectation, so it depends only on the weak order. Both step factors have mean $1 + r\Delta t$, which gives the exact identity $E[S_T^{\Delta t}] - E[S_T] = S_0(1 + r\Delta t)^m - S_0e^{rT}$ ($m = T/\Delta t$), an analytic check on the simulation.

2.5 Importance Sampling

Importance sampling simulates under a shifted drift $\mu = r + \sigma\theta$ and corrects with a likelihood ratio. Writing $\tilde{W}$ for the Brownian motion built from the simulated normals, the risk-neutral Brownian motion is $W = \tilde{W} + \theta t$, and by Girsanov’s theorem the likelihood ratio telescopes to a function of the terminal value alone:

$$L = \exp\left(-\theta\tilde{W}_T - \frac{1}{2}\theta^2 T\right). \quad (8)$$

The estimator is the sample mean of $L \cdot Y$, unbiased for any fixed θ . The drift is chosen by a heuristic that places the expected terminal log-price on the barrier,

$$\theta_0 = \frac{1}{\sigma} \left[\frac{\ln(B/S_0)}{T} - \left(r - \frac{1}{2}\sigma^2\right) \right], \quad (9)$$

followed by a 7-point pilot search over $\theta_0 \cdot \{0, 0.25, \dots, 1.5\}$ on an independent 10,000-path pilot, minimizing the empirical second moment of the weighted payoff. Every estimate reports the Kish effective sample size [10], $\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$, and the maximum weight, which together diagnose whether the result rests on a handful of paths.

Table 2: Frozen scenarios used throughout the experiments.

Scenario	Type	Barrier B
paper_barrier	Up-and-in call	110.6772
mid_barrier	Up-and-in call (parity check)	130
deep_barrier	Up-and-in call	140
rare_barrier	Up-and-in call (extension)	160
paper_asian_geo	Geometric Asian call/put	—
asian_30d	Geometric Asian, last 30 days	—
asian_arith	Arithmetic Asian call (no closed form)	—

Table 3: 252-date (daily-monitoring) reference prices used for every barrier RMSE, against the continuous Reiner–Rubinstein and BGK values.

Barrier	252-date reference	Continuous	BGK
$B = 110.68$	7.094133 ± 0.000096	7.105528	7.092995
$B = 140$	3.385730 ± 0.001014	3.545321	3.377810

2.6 Benchmarking and Acceptance Criteria

Every estimator is validated against four criteria fixed before any experiment was run: (1) within 3 standard errors of a reference price wherever one exists; (2) 95% confidence-interval coverage across replicates; (3) every speedup reported as efficiency = $1/(\text{variance} \times \text{time})$ with a bootstrap confidence interval [5] (2,000 resamples), since with $R = 20$ replicates a variance estimate alone carries about $\pm 32\%$ relative error; and (4) error reported as RMSE, not variance alone, since a biased low-variance estimator would otherwise look artificially good. RQMC has no valid within-run standard error, so its coverage is instead checked with a t -interval over its R independent scrambles.

3 Implementation and Experiments

3.1 Scenarios and Protocol

All scenarios share $S_0=100$, $r=0.03$, $\sigma=0.2$, $T=1$ year, $K=105$, 252 daily monitoring dates, sample grid $N \in \{256, 1024, 4096, 16384, 65536\}$ (powers of two, for QMC), and $R = 20$ independent replications per cell, all seeded from one deterministic base seed (Table 2).

3.2 Reference Prices

The Asian references are closed forms. For the barriers, the continuous Reiner–Rubinstein formula would conflate the discrete-monitoring bias with sampling error once a variance-reduced estimator’s error drops low enough, so every barrier RMSE is instead measured against a dedicated 252-date reference built from 64 independent bridge scrambles at $N = 65,536$, on a seed stream separate from every sweep (Table 3). The paper-barrier reference agrees with the in-out-parity value of the baseline replication (7.0941 ± 0.0002 , 0.15 combined SE) and with the control-variate value (7.093583 , 1.86 combined SE). The arithmetic Asian’s reference (3.162981 ± 0.000064) is built the same way.

3.3 Baseline Replication

We first reproduce the base paper’s own setup: the barrier call under all three discretization schemes (paired by sharing random numbers across schemes), and the geometric Asian call, put, and 30-day-window call under the exact scheme, each with plain MC and antithetic variates. Two high-precision reference runs validate the sampling: a 4,000,000-path in-out-parity computation of the 252-date barrier price, and a 1,000,000-path common-random-number comparison of the three discretization schemes against the exact scheme.

3.4 Discretization-Order Experiment

To explain the base paper’s scheme finding, a dedicated experiment measures the strong and weak order of Euler–Maruyama and Milstein directly (Section 2.4) for $\Delta t = 1/4, \dots, 1/512$, using one shared 512-step Brownian path per sample (coarsened to each step size) so every scheme and step size sees the same underlying randomness. Errors are taken against the exact scheme on the same grid, which isolates the scheme error from the barrier’s monitoring bias ($20 \times 65,536$ paths per step size).

3.5 Control Variates and the Arithmetic Asian

The arithmetic-average payoff is the mean of the 252 simulated monitoring prices, run at every N in the grid with $R = 20$ replications, using the discrete geometric Asian as the control (Section 2.2). Since the arithmetic Asian has no closed form, an independent 2,000,000-path plain-MC run provides an external validation value. The same control-variate machinery, with the terminal European call as control, is applied to the paper barrier and to a deep barrier ($B = 140$) to contrast a near-perfect control with a weak one. A separate sweep at 12, 52, and 252 monitoring dates, with the Broadie–Glasserman–Kou continuity correction [3] applied for comparison, isolates the discrete-monitoring bias.

3.6 Quasi-Monte Carlo Experiments

Before pricing any option, the RQMC implementation is validated on two integrals with known closed-form answers: a smooth 8-dimensional exponential integral and a discontinuous 2-dimensional Gaussian orthant probability (32 scrambles, $N = 2^6 \dots 2^{14}$). RQMC’s RMSE slopes are -0.83 and -0.77 , against -0.43 and -0.50 for plain MC, with RMSE ratios of $29\times$ and $18\times$ at $N = 16,384$. The main sweep compares plain MC, bridge-constructed RQMC, and incremental-constructed RQMC on the geometric Asian call and on the paper and deep barriers ($R = 20$ scrambles per cell), with barrier RMSE measured against the 252-date references of Section 3.2.

3.7 Importance Sampling Experiments

Before any option is priced, the likelihood-ratio machinery is validated by estimating a known tail probability, $P(Z > 4)$, under a shifted proposal, and checking it against the exact value. The main sweep compares plain MC, antithetic variates, and importance sampling on the paper barrier, the deep barrier, and a runtime-only extension scenario, *rare_barrier* ($B = 160$), added because the deep barrier’s 9.3% knock-in rate turned out not to be genuinely rare. A fixed- θ scan over $\theta_0 \cdot [0, 3]$ with common random numbers

Table 4: Reproduction of the base paper’s two central findings.

Base-paper finding	Our reproduction
Antithetic variates accelerate convergence $\approx 1.5\times$ (barrier), $\approx 1.3\times$ (Asian)	Reproduced. Variance ratio at equal path count $1.47\times$ (barrier) and $1.35\times$ (Asian call), stable across the whole N grid ($1.44\text{--}1.48$; $1.34\text{--}1.53$); predicted by $1/(1 + \rho)$
Discretization scheme has a negligible effect	Reproduced. Price difference from the exact scheme -0.00057 (Euler–Maruyama) and -0.00105 (Milstein) at daily steps, about $50\times$ below the plain-MC standard error at $N = 65,536$; explained by weak order 1 for both schemes

across θ traces the full per-path-variance curve that the 7-point pilot search samples from.

3.8 Master Comparison

A final experiment benchmarks all five methods together, on one machine, single-threaded, so that the timings behind the efficiency metric are directly comparable. The sweep covers 17 (scenario, method) cells $\times$ the N grid $\times R = 20$: both barriers with all five methods; the arithmetic Asian with plain, antithetic, control variate, and RQMC; and the geometric Asian with plain, antithetic, and RQMC (the control variate is not applicable to an option that already has a closed form, and importance sampling targets barrier crossings specifically). Every runtime includes all of a method’s own overhead: control-variate and importance-sampling pilot runs, and RQMC’s Sobol scramble. An initial attempt ran each method as one uninterrupted block and found machine-speed drift in its first two minutes inflating one scenario’s timings relative to another’s by up to $2\times$ for identical work; the final sweep instead *interleaves* all 17 cells in a freshly shuffled order within each (replicate, N) block, after an untimed warm-up pass. All 1,700 prices from the interleaved run are bit-identical to the blocked run, confirming that only the timing changed.

4 Results and Discussion

4.1 Baseline Validation and Reproduction of the Base Paper’s Findings

Every baseline estimator lands within 3 standard errors of its closed-form reference in 98–100% of replicates, and 95% confidence-interval coverage, pooled over all five N , is 91–98%, meeting both acceptance criteria from Section 2.6. Figure 1 shows the estimated price and confidence interval against N for the barrier call. Table 4 summarizes our reproduction of the base paper’s two central findings (Section 1.3); both hold, and the rest of this subsection explains why.

Why the antithetic gain is $1.5\times$ and $1.3\times$. With N paths arranged as $N/2$ antithetic pairs, the estimator’s variance is $\sigma_Y^2(1 + \rho)/N$, where $\rho = \text{Corr}(Y, Y')$ is the correlation between a path’s discounted payoff and its mirror’s, so the variance ratio against plain

MC at equal path count is exactly $1/(1 + \rho)$. The measured pair correlations predict the measured ratios to within the rounding of ρ : $\rho = -0.32 \Rightarrow 1.47\times$ (barrier), $\rho = -0.26 \Rightarrow 1.35\times$ (geometric Asian call), and $\rho = -0.32 \Rightarrow 1.48\times$ for the 30-day-window call. The same relation explains the Asian *put*: at $K = 105 > S_0$ the put is usually in the money, so its payoff is close to linear (odd) in the driving normals, the pair correlation reaches $\rho = -0.74$, and the gain rises to $3.91\times$. The relation also predicts where antithetic variates help less. As the payoff becomes less linear in Z , ρ moves toward 0: the within-run ratio falls from $1.5\times$ on the paper barrier to $1.1\times$ on the deep barrier ($B = 140$) and $\approx 1.0\times$ on the rare barrier ($B = 160$), and it is $1.4\times$ on the arithmetic Asian. In the single-machine master sweep (Section 4.6) antithetic costs about the same per path as plain MC on the paper barrier and the Asians (time ratio $0.94\text{--}0.99$; 0.73 on the deep barrier), so there its efficiency gain is essentially its variance ratio. These ratios are measured within each run from $N/2$ independent pairs; the across-replicate ratio at $R = 20$ has a sampling range of roughly $[0.4, 2.5]\times$, too wide to distinguish 1.3 from 1.5 , which is why we report the within-run value.

Why the scheme choice is negligible. Measured directly with common random numbers (Section 3.4), Euler–Maruyama has strong order 0.49 and Milstein 0.97 , matching theory (Figure 3). For a price, however, only the weak order matters, and both schemes have weak order ≈ 1 on both the vanilla call and the barrier (Figure 4): $1.01/0.99$ on the call and $1.03/0.99$ on the barrier (Euler–Maruyama/Milstein). The simulated $E[S_T]$ error matches the exact identity of Section 2.4 within 3 SE at all eight step sizes. At $\Delta t = 1/256$, Milstein’s path-wise error is $66\times$ smaller than Euler–Maruyama’s (0.0022 vs. 0.145), yet the two barrier price biases are of the same size (-0.00104 and -0.00078), and both are $47\times$ smaller than the plain-MC standard error at $N = 65,536$ and $171\times$ smaller at $N = 5,000$. At any practical sample size, then, the three schemes produce visually indistinguishable price estimates (Figure 2), exactly as the base paper reports; our order study shows that this is a consequence of weak-order theory and will carry over to other payoffs under GBM at daily steps. A practical corollary is that under GBM the exact log-normal step, which has *zero* discretization error, costs about the same as either scheme (0.64 s, against 0.54 s for Euler–Maruyama and 0.77 s for Milstein per $65,536$ paths at $m = 512$), so we use it for every later experiment.

4.2 Additional Baseline Findings

Convergence rate and the cost of precision. Across all eight baseline scenario/method pairs, RMSE falls with fitted slopes of -0.45 to -0.52 , consistent with the theoretical $N^{-1/2}$ (Figure 5), and $\text{RMSE} \cdot \sqrt{N}$ stays roughly constant over the grid ($9.5\text{--}14.3$ for the barrier, $3.9\text{--}6.8$ for the Asian call): the error keeps shrinking at the same rate, with no threshold beyond which extra paths stop helping. The practical consequence is the cost of precision. With per-path standard deviations of 12.55 (barrier) and 5.86 (Asian call), the 95% confidence interval is still $\pm 4.9\%$ of the barrier price at $N = 5,000$ and $\pm 12\%$ of the Asian price at $N = 1,000$, and reaching $\pm 1\%$ needs about 1.2×10^5 and 1.5×10^5 paths respectively. Confidence-interval plots on a logarithmic N axis look settled after a few thousand paths, but each further halving of the error still costs four times

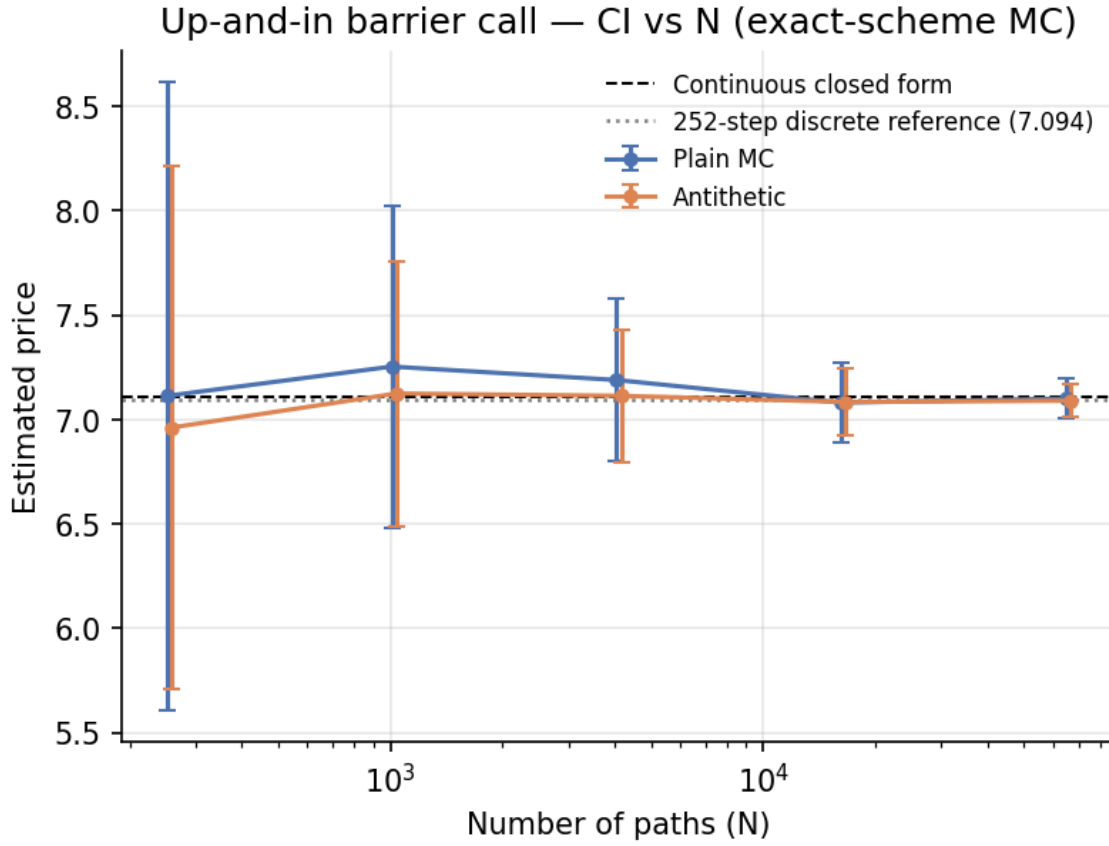


Figure 1: Baseline replication: estimated price $\pm$ 95% CI vs. N for the up-and-in barrier call, plain MC and antithetic variates, against the continuous closed form (dashed) and the 252-date reference (dotted).

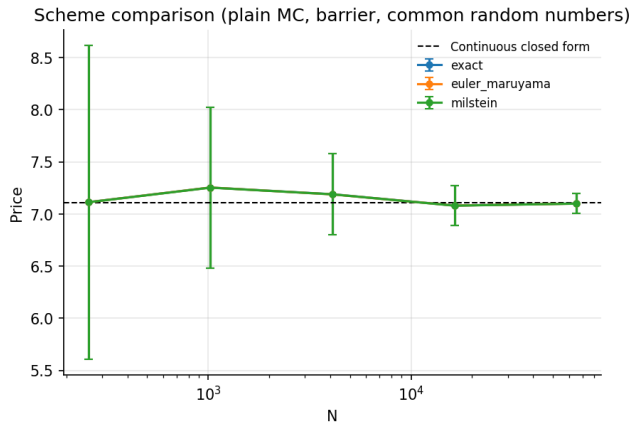


Figure 2: Barrier price $\pm$ 95% CI vs. N under the exact, Euler-Maruyama, and Milstein schemes with common random numbers: the three curves coincide at every N .

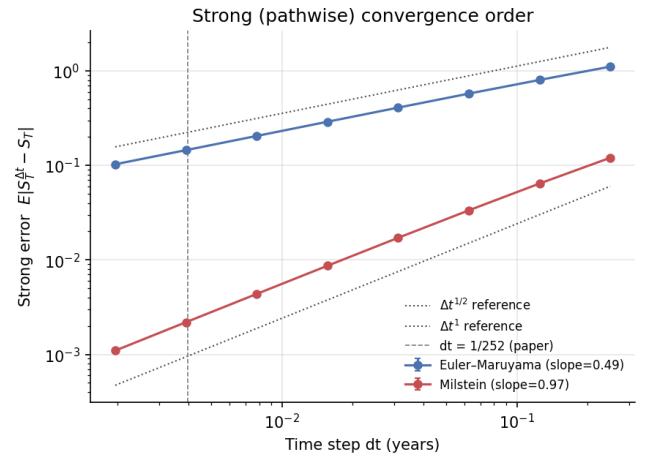


Figure 3: Strong-order convergence of Euler-Maruyama and Milstein: fitted slopes 0.49 and 0.97, matching the theoretical 0.5 and 1.0.

the work; this is the wall that the variance-reduction techniques of the following sections attack.



Figure 4: Weak-order convergence of both schemes for the terminal price and the barrier option: both close to the theoretical order 1, and indistinguishable from each other at practical sample sizes.

Sensitivity to the averaging window. The base paper also prices a geometric Asian call averaged over the last 30 days. Its discrete closed form is the same formula with a different averaging grid, and the price is highly sensitive to that grid: 6.708 for a 30-date window against 3.000 for the full 252-date window (vanilla call: 7.128). Our antithetic estimate at $N = 65,536$, 6.698 ± 0.0066 , validates the 30-date value. The mechanism is that a shorter window averages over less of the path, so the volatility of the average approaches that of S_T and the price moves toward the vanilla call; reading “30 days” as about 21 trading days gives ≈ 6.84 , closer still. The antithetic gain changes only modestly ($1.48\times$ vs. $1.35\times$, both predicted by $1/(1 + \rho)$), so the window affects the price far more than it affects the sampling method.

The discrete-monitoring bias. Our initial estimate of the 252-step discretely monitored barrier price was ≈ 7.076 (bias ≈ -0.03). A high-precision in-out-parity computation (SE 0.0002) instead gives 7.0941, a bias of only -0.0114 against the continuous formula, confirmed independently by the control-variate estimator (Section 4.3) and by the 64-scramble RQMC reference (Table 3). At $N = 1,024$, a single plain-MC run’s standard error (≈ 0.39) is about $34\times$ larger than this bias, so plain Monte Carlo cannot see it at any practical sample size. The bias grows sharply with the barrier level: on the deep barrier it is -0.160 (4.5% of the price), $14\times$ the paper barrier’s. Only paths that cross B and then finish below it can be affected by a missed crossing, since a path finishing above B is caught on the last date; that exposed part is 33% of the deep barrier’s continuous value but only 3% of the paper barrier’s. The BGK correction over-corrects both, landing about 8 SE below the deep-barrier reference. Any deep-barrier RMSE taken against the continuous formula would be dominated by this 0.16 bias, which exceeds the error of every variance-reduced method; this is why every RMSE in this report is measured against the 252-date references.

4.3 Control Variates: Arithmetic Asian and the Barrier Collapse

At $N = 65,536$, the control-variate estimate of the arithmetic Asian call is 3.162776 ± 0.000361 (95% CI of the replicated mean), with correlation $\rho = 0.999488$ against its geometric-Asian control and an observed within-run variance-reduction factor of $975.7\times$, matching the theoretical $1/(1 - \rho^2) = 976.6\times$ almost exactly. An independent 2,000,000-path plain-MC run gives 3.162517 ± 0.008511 , within 0.06 combined standard errors — validating the arithmetic-Asian price without a closed form to check against. Figure 6 shows the observed

Table 5: RMSE-vs- N slopes against the 252-date references, with 2,000-resample bootstrap 95% CIs (QMC experiment, $R = 20$).

Scenario	Plain MC	RQMC, bridge	RQMC, incremental
Geometric Asian	−0.57	−0.72 [−0.78, −0.65]	−0.61
Paper barrier	−0.52	−0.80 [−0.88, −0.71]	−0.76
Deep barrier	−0.44	−0.60 [−0.67, −0.52]	−0.58

VRF tracking the theoretical value at every N , and Figure 7 confirms the $1/(1 - \rho^2)$ relationship across every scenario and sample size tested.

The same control (a terminal European call) behaves very differently on two barriers. On the paper barrier ($B = 110.68$), where the up-and-out leg is worth only about 0.3% of the price, the up-and-in payoff is almost identical to a vanilla call path by path, giving $\rho = 0.9997$ and a VRF of $1542\times$. On the deep barrier ($B = 140$), many in-the-money vanilla-call paths never knock in, so the correlation collapses to $\rho = 0.830$ and the VRF falls to only $3.2\times$ (Figure 8). This is the central pedagogical result of this technique: a control variate’s value is not intrinsic, but depends entirely on how closely it tracks the specific target.

The same estimator, applied with 12, 52, and 252 monitoring dates, resolves the discrete-monitoring bias directly: at 252 dates it measures -0.011945 , matching Section 4.1’s reference to within 0.0006, and resolves it at 54.1 standard errors of the replicated mean. The BGK correction is accurate to within 0.0006–0.0011 at 252 dates but over-corrects substantially at coarser monitoring (Figure 9), confirming it is an asymptotic approximation rather than an exact discrete-barrier formula.

4.4 Quasi-Monte Carlo Convergence

Measured against the 252-date reference, bridge-constructed RQMC achieves RMSE-vs- N slopes of -0.80 on the paper barrier, -0.60 on the deep barrier, and -0.72 on the geometric Asian call, all steeper than plain MC’s ≈ -0.5 (Figures 10 and 12). Measuring the paper barrier against the *continuous* formula instead would have given a misleadingly flat slope of -0.26 , since the -0.011 monitoring bias would floor the apparent error once RQMC’s own error dropped below it — a direct illustration of why the reference price must match the simulation’s monitoring frequency. At $N = 65,536$ in this experiment’s own run, RQMC’s variance-reduction factor reaches $6988\times$ on the paper barrier, $1052\times$ on the geometric Asian, and $45.5\times$ on the deep barrier (the independent master sweep of Section 4.6 gives $5723\times$ and $40\times$ on the two barriers, well within their F -distribution confidence intervals); the barrier’s large gain mirrors the control variate’s $1542\times$ for the same reason, that the option is nearly a one-dimensional function of the terminal price. The Brownian bridge construction is responsible for nearly all of this gain: it concentrates 100% of $\text{Var}(\log S_T)$ and 75–98% of $\text{Var}(\log G)$ (the Asian average) into the first few Sobol dimensions, where the sequence is most uniform (Figure 11), giving a further 27 – $536\times$ reduction in variance over the naive incremental construction (Figure 13). Table 5 gives every slope with its bootstrap 95% CI; every plain-MC slope is consistent with -0.5 .

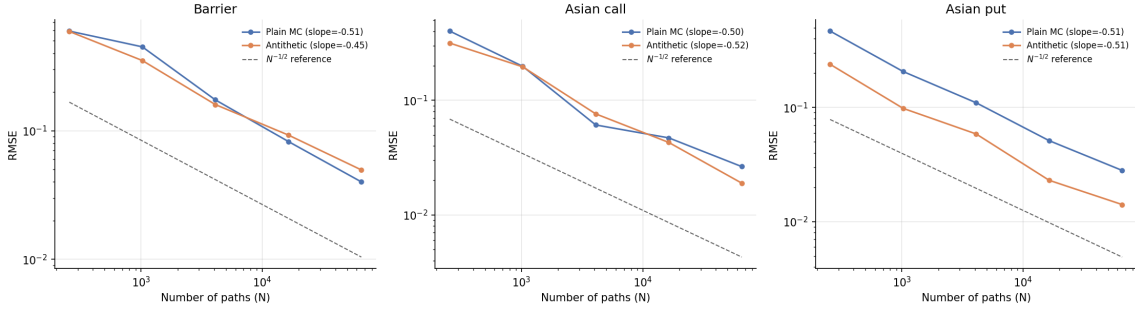


Figure 5: RMSE vs. N on log-log axes for the baseline scenarios, with fitted slopes close to the theoretical $N^{-1/2}$ rate.

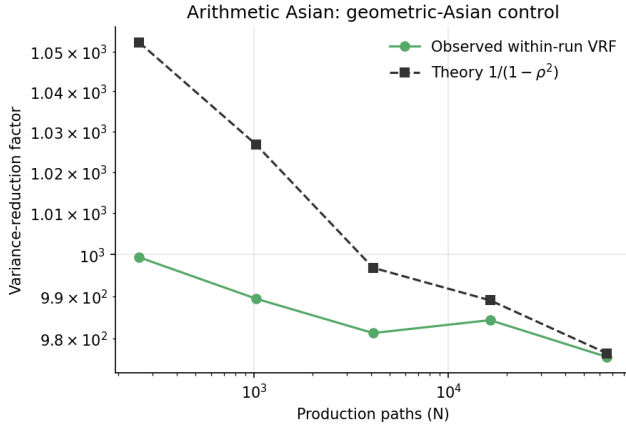


Figure 6: Control-variate variance-reduction factor for the arithmetic Asian call: observed vs. theoretical $1/(1-\rho^2)$, stable at $\approx 10^3$ across the sample grid.

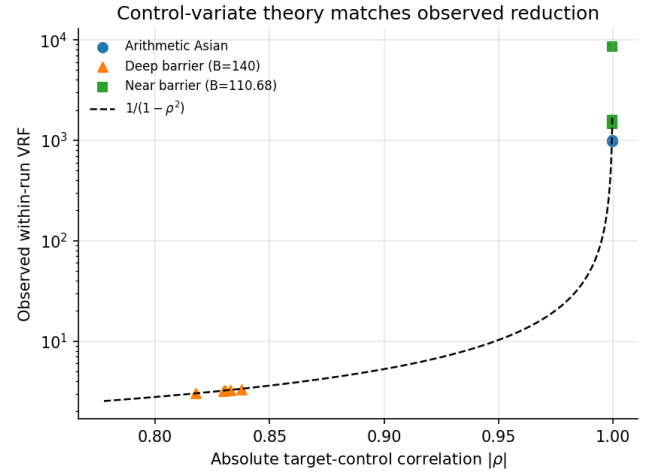


Figure 7: Observed variance-reduction factor vs. correlation ρ , compared with the theoretical curve $1/(1-\rho^2)$, across all tested scenarios.

Why the rates differ. We expected RQMC to help barriers less than Asians, because a barrier's payoff is discontinuous. The mechanism holds, but the grouping does not. The *deep* barrier behaves as predicted (-0.60 , VRF $45\times$): its knock-in depends on the path maximum, a jump spread over many bridge coordinates whose boundary is not axis-aligned, the same effect the orthant smoke test shows in two dimensions. The *paper* barrier is the exception (-0.80 , the largest gain in the study): at $B = 110.68$ the up-and-out leg is worth only 0.023 , so the payoff is almost exactly $(S_T - K)^+$, and under the bridge S_T depends on Sobol dimension 0 alone. The integrand is effectively one-dimensional with a single kink, QMC's best case. This contradicted our expectation, so we investigated before reporting it: the RQMC mean agrees with two independent references, the bridge passes its orthogonality tests, and the effect matches the control variate's $1542\times$ on the same option. The *Asian* slope, -0.72 , falls short of the predicted -0.9 : the bridge puts 98.4% of $\text{Var}(\log G)$ in the first four dimensions, but the remaining 1.6% is spread thinly over hundreds of poorly balanced high-index dimensions and decays almost at the MC rate, and the payoff kink slows the fast part too.

Fixed cost and efficiency. Scrambling a 252-dimensional Sobol generator costs about 35 ms per estimate, against 3–4 ms for an entire plain-MC run at $N = 256$. At $N = 256$ on the deep barrier RQMC is therefore *less* efficient than plain MC ($0.4\times$ here, $0.45\times$ in the master sweep), and it pays off only from $N \approx 1,024$. Because its error also falls faster, its advantage grows with N : at $N = 65,536$ the bootstrap efficiency ratios in this run are 621 [303, 1372] (geometric Asian), 3768 [1729, 8954] (paper barrier) and 24.7 [8.6, 62.3] (deep barrier), with RQMC costing 1.7–1.85 $\times$ plain MC per estimate (timings indicative; the master sweep is authoritative).

Validation. Since RQMC has no per-run CI, each cell is checked with a t -interval over its 20 scrambles: all 45 cells of all three methods lie within 3 SE of the reference, and the 95% t -interval covers it in 43 of 45 cells.

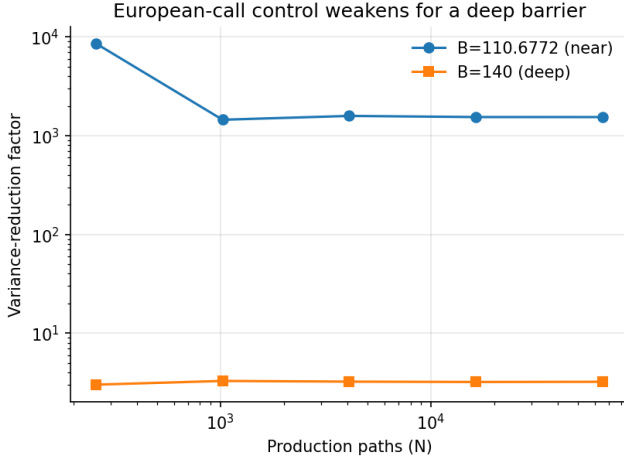


Figure 8: The vanilla-call control variate on the paper barrier (near-perfect, VRF $\approx 1542\times$) versus the deep barrier (weak, VRF $\approx 3.2\times$).

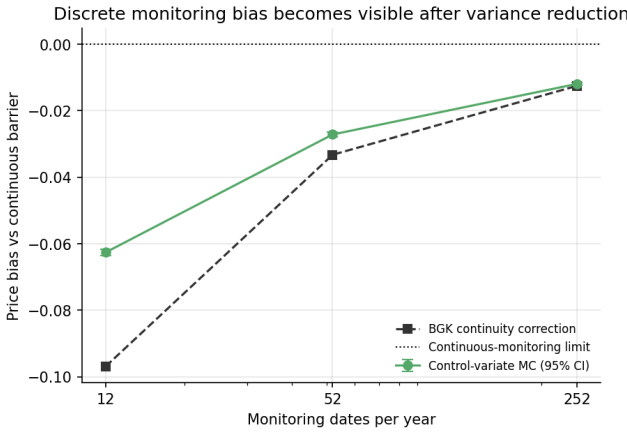


Figure 9: Discrete-monitoring bias vs. number of monitoring dates, measured by the control-variate estimator, against the BGK continuity correction.

4.5 Importance Sampling for Rare Barrier Crossings

Table 6 summarizes importance sampling at $N = 65,536$ on three barriers of increasing rarity: the paper barrier (60% knock-in probability), the deep barrier (9.3%), and the extension scenario rare_barrier at $B = 160$ (1.9%).

The pattern matches theory exactly: importance sampling’s advantage grows as the target event becomes rarer. Figure 14 shows the deep-barrier price converging correctly alongside its efficiency gain over plain MC across the sample grid; the gain is only visible from $N \approx 2,000$ –4,000 upward, since the 10,000-path pilot’s fixed cost dominates at smaller N . Figure 15 shows the likelihood-ratio weight distribution at the tuned drift for all three barriers: although the rare barrier’s effective sample size is only 1.1% of N , the largest

Table 6: Importance sampling at $N = 65,536$: within-run variance-reduction factor and efficiency vs. plain MC. Paper and deep barriers: master sweep; rare barrier: the importance-sampling experiment (indicative timing).

Barrier	Knock-in %	VRF	Efficiency
Paper ($B = 110.68$)	60%	$4.6\times$	$2.8\times$
Deep ($B = 140$)	9.3%	$9.8\times$	$8.6\times$
Rare ($B = 160$, extension)	1.9%	$22.2\times$	$\approx 49\times$

weights fall almost entirely on paths with zero payoff, so the price estimate is not dominated by a handful of paths — the top 1% of paths carry only 6% of the price even in that case.

4.6 Master Comparison of All Five Methods

Table 7 gives the headline result of the project: efficiency relative to plain Monte Carlo at $N = 65,536$, from the single authoritative, interleaved run (Section 3.8), with bootstrap 95% confidence intervals.

Figure 16 shows the same comparison graphically, and Figure 17 shows the RMSE-vs- N curves for every method on every scenario, from which the headline efficiency numbers are derived.

Three findings stand out. First, antithetic variates deliver the gain measured in Section 4.1 — within-run variance ratios of 1.4–1.5 $\times$ on the paper barrier and both Asians at no extra cost per path — but a gain of that size is smaller than the $\pm 32\%$ sampling noise of a 20-replicate variance estimate, so the across-replicate efficiency intervals in Table 7 all include 1 $\times$. The techniques that exploit the payoff’s structure reach two to three orders of magnitude more. Second, the control variate and RQMC are *statistically tied* on the paper barrier and the arithmetic Asian, because they exploit the same underlying structure (the payoff is nearly a function of one Gaussian), while RQMC pulls ahead as N grows because it improves the convergence *rate* rather than just the constant. Third, the deep barrier is hard for every technique, since its payoff depends on the path maximum rather than a single coordinate; importance sampling is the only technique whose advantage grows as the knock-in becomes rarer still (Table 6).

Converting efficiency into a practical number: reaching one-cent accuracy ($\epsilon = 0.01$) on the arithmetic Asian needs about 270,000 plain-MC paths (2.3 s) but only 340 control-variate paths (5 ms), roughly 500 $\times$ faster; on the deep barrier at $\epsilon = 0.001$, plain MC would need an extrapolated 6.9×10^8 paths (almost two hours), while RQMC needs about 30 s.

5 Conclusion and Future Work

5.1 Conclusion

We rebuilt the base paper’s experiment from scratch, validated every estimator against closed forms, and reproduced its two central findings. Its antithetic speedups (1.47 $\times$ barrier, 1.35 $\times$ Asian call) are confirmed as variance ratios and are predicted by the pair correlation through $1/(1 + \rho)$, which also explains the larger gain on the in-the-money Asian put and the smaller gain on rarer barriers. Its conclusion that the discretization scheme has a negligible effect is confirmed and explained by weak-order theory: Milstein’s

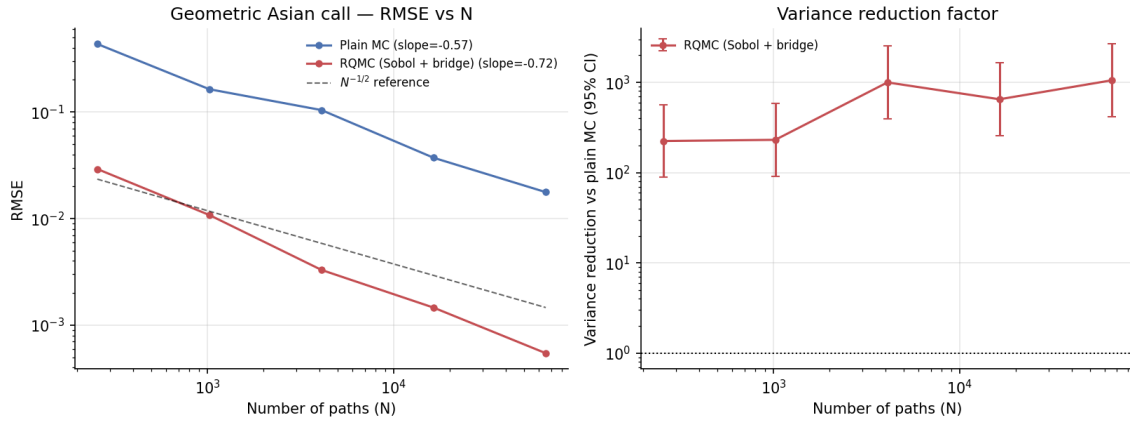


Figure 10: Geometric Asian call: RMSE vs. N for plain MC and RQMC with a Brownian bridge (left), and RQMC’s variance-reduction factor with F -distribution 95% CIs (right).

Table 7: Master comparison: efficiency relative to plain Monte Carlo at $N = 65,536$, with 95% bootstrap confidence intervals. “n/a” marks a method not applicable to that option.

Method	Paper barrier	Deep barrier	Arithmetic Asian	Geometric Asian
Plain MC	1	1	1	1
Antithetic	1.2 [0.5, 2.8]	1.1 [0.5, 2.9]	0.97 [0.35, 2.6]	1.1 [0.5, 2.9]
Control variate	1,602 [656, 4,276]	4.2 [1.6, 10.0]	770 [266, 1,970]	n/a
RQMC + bridge	2,260 [1,065, 5,050]	21.3 [7.9, 51.4]	794 [269, 2,120]	791 [358, 1,638]
Importance sampling	2.8 [1.3, 5.7]	8.6 [3.0, 37.8]	n/a	n/a

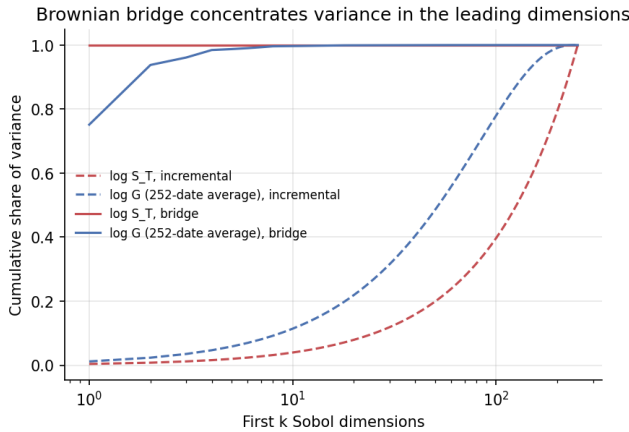


Figure 11: Share of the variance of $\log S_T$ and of the log geometric average carried by the first k Sobol dimensions, under the Brownian bridge (solid) and the incremental construction (dashed). Exact calculation from the construction matrix, not a simulation.

much better pathwise accuracy does not reach the price, where both schemes are first order and their bias is far below the sampling error. Along the way we quantified how slowly plain Monte Carlo buys precision (about 10^5 paths for $\pm 1\%$) and how strongly the Asian

price depends on its averaging window. We then implemented three advanced variance-reduction techniques, the direction the base paper recommends for future work, and benchmarked all five methods on one efficiency metric from a single authoritative run. Control variates and randomized quasi-Monte Carlo are 10^2 – 10^3 times more efficient than plain Monte Carlo wherever a payoff is nearly a function of one Gaussian; importance sampling is the specialist for genuinely rare barrier crossings; and antithetic variates remain a simple, model-free first step whose modest 1.1–1.5 $\times$ gain depends on how linear the payoff is. Variance reduction additionally makes a real, systematic discrete-monitoring bias visible that plain Monte Carlo is too noisy to detect. **Our recommendation:** to reach a target accuracy on a barrier option, use randomized quasi-Monte Carlo with a Brownian bridge; on the arithmetic-average Asian option, use the geometric-Asian control variate.

5.2 Limitations

With $R = 20$ replicates, efficiency confidence intervals span a factor of 3–5, so techniques within about $2\times$ of each other (control variate vs. RQMC on two scenarios; RQMC vs. IS on the deep barrier) cannot be statistically ranked. The authoritative timings come from one machine, single-threaded; cost ratios should transfer across machines, but absolute numbers will not. Our importance-sampling drift search uses a fixed 7-point grid tied to a heuristic starting point, which under-reaches the true optimum on the paper barrier by about $2.3\times$. Both the control-variate and importance-sampling

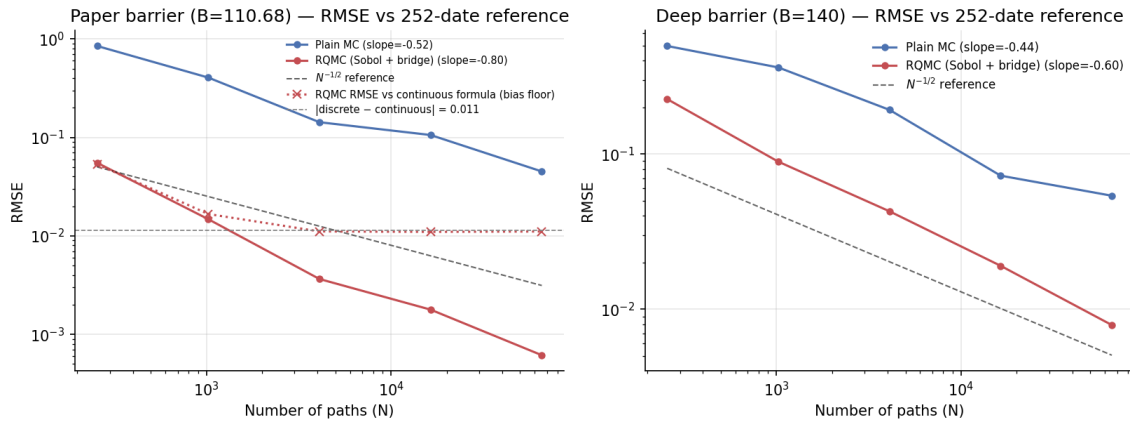


Figure 12: RQMC versus plain Monte Carlo on the paper and deep barriers, RMSE measured against the 252-date reference.

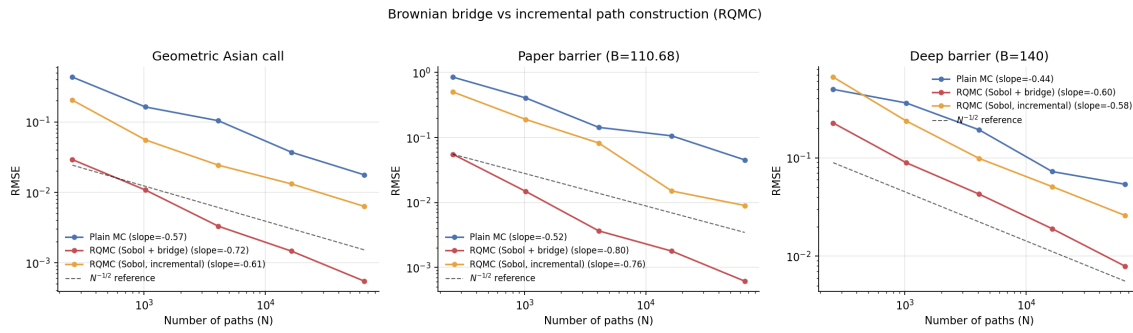


Figure 13: Brownian-bridge versus incremental Sobol-dimension construction, across the geometric Asian call, the paper barrier, and the deep barrier.

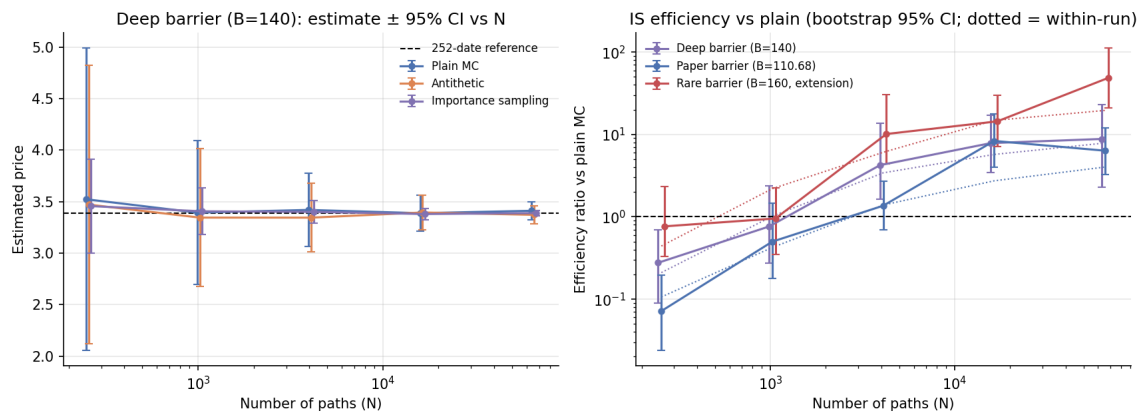


Figure 14: Importance sampling on the deep barrier: price convergence (left) and efficiency vs. plain MC (right) across the sample grid.

estimators are validated only under the exact GBM scheme; their biases under Euler–Maruyama or Milstein were not characterized. Finally, our replication fixes the implementation details the base

paper leaves open (step count, random seed, Asian strike and averaging grid) at the most plausible values; the findings of Section 4.1 do not depend on these choices among the alternatives we tried (e.g. a 21-trading-day reading of the 30-day window).

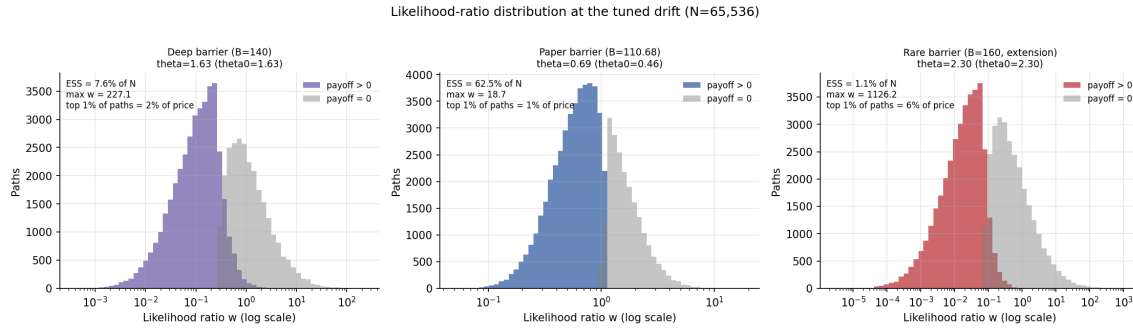


Figure 15: Likelihood-ratio weight distributions at the tuned drift, for the deep barrier, the paper barrier, and the rare extension barrier.

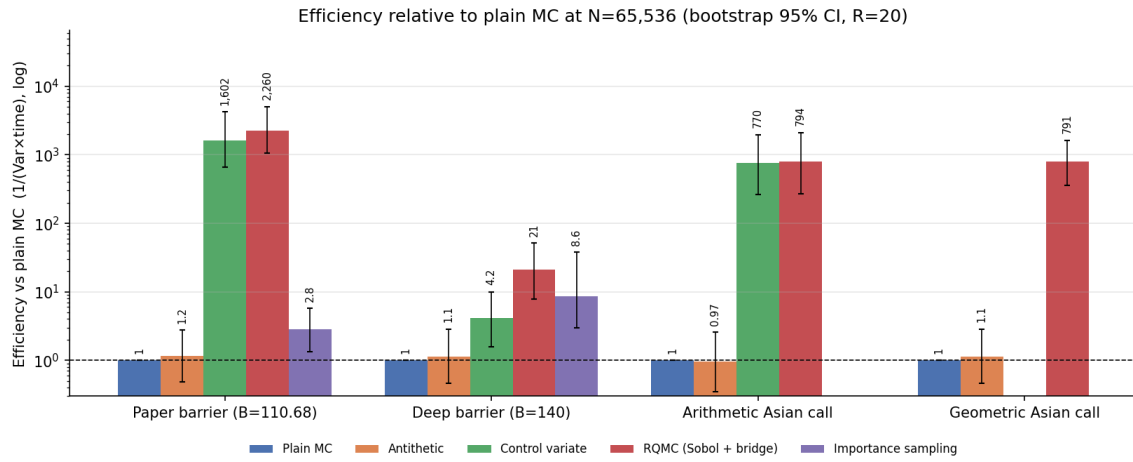


Figure 16: Efficiency vs. plain Monte Carlo across all five methods and all four scenarios, at $N = 65,536$ (log scale).

5.3 Future Work

Natural next steps are: (1) combining techniques, e.g. RQMC with a control variate or with importance sampling, which should dominate on the deep barrier where each alone gains only 9–21 $\times$; (2) better importance-sampling drift tuning via cross-entropy or large-deviations methods; (3) multilevel Monte Carlo to address discretization error, the dual problem to the sampling-error reduction studied here; (4) richer models (stochastic volatility, double barriers) and products (Greeks), where variance reduction matters even more; and (5) a larger replicate count for the master sweep, to resolve the near-ties identified above.

Author Contributions

The project ran as a five-stage relay, with each member owning one stage end to end (code, tests, experiments, and the corresponding report content); Stage 6 (assembly) was shared. Nafis Nahian (2105007) built the simulation engine, estimators, and benchmarking infrastructure, and ran the baseline replication. Aritra Debnath (2105010) implemented control variates and the arithmetic-Asian and barrier-bias experiments. Zaki Rehnoom Unmona (2105016) implemented

quasi-Monte Carlo, the Brownian bridge, and the discretization-order study. Tamzeed Mahfuz (2105012) implemented importance sampling, the master comparison, and the final synthesis. Arnob Biswas (2105015) built the analytic closed-form oracles and the foundational configuration and testing infrastructure that every later stage imports.

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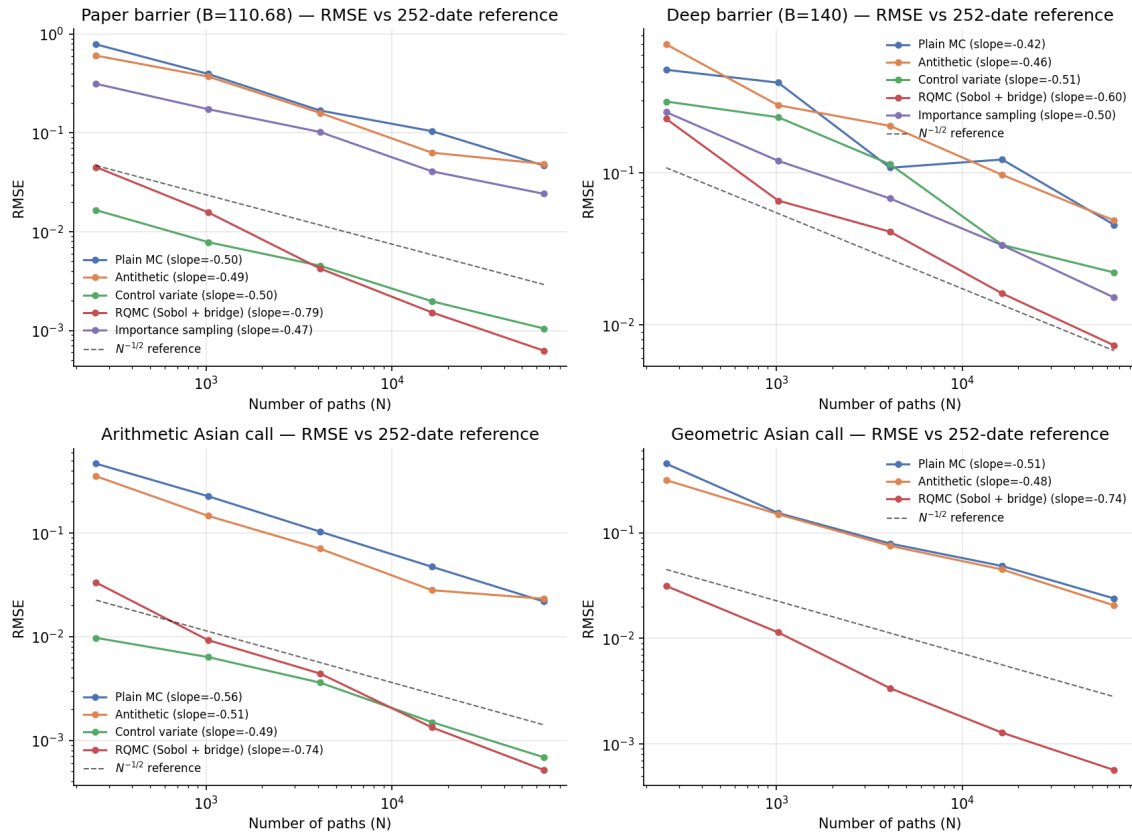


Figure 17: RMSE vs. N for every method on every scenario, from the master sweep (log-log axes).

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A Reproducing Every Number

Every number reported in this paper is generated by the code in the GitHub repository linked after the title. From a fresh clone:

```
pip install -r requirements.txt
pytest tests/ -q # 90 tests
python run_all.py --analyze-only # regenerate every figure
# and table from the committed
# raw data (about 2 minutes)
python run_all.py # full re-simulation from
```

scratch (about 25 minutes,
single-threaded)

Every price in the repository is produced by a deterministic random seed keyed on the scenario, method, sample size, and replicate index, so a re-simulation reproduces every price exactly on the same library versions. Timings, and therefore efficiency numbers, depend on the machine; the authoritative timings are those of the master sweep (Section 3.8).